

Excitonic insulators

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1 Introduction

Often, excitonic insulators are described in a BCS-like theory by using the mean-field (MF) approximation. In this framework, one introduces an order parameter as the expectation value of the operator which creates interband excitons. A non-zero order parameter opens a gap in the effective band structure, producing an insulating state. This gives a correct description of the semimetal-insulator transition governed by electron-hole interaction. If we employ the MF approximation in connection to the linear response theory, we describe the quasiparticle transport determined by an assumed band structure. However, especially due to neutrality of excitons it is unclear to which extent this approach is valid. To clarify this issue, the goal of this project is to study an excitonic model with various methods, one of them being the MF, which I discuss presently.

2 Mean-field description

We consider a one-dimensional system with two orbitals per site at half filling described by the Hamiltonian

$$H = \sum_i \left[(t_1 c_{1,i}^\dagger c_{1,i+1} - t_2 c_{2,i}^\dagger c_{2,i+1} + \text{H.c.}) - \varepsilon n_{1,i} + \varepsilon' n_{2,i} + V n_{1,i} n_{2,i} \right]. \quad (1)$$

Here, $t_1, t_2 > 0$ are nearest-neighbor hopping amplitudes and $\pm\varepsilon$ are onsite energies. We choose $t_1 = 1$ and $t_2 = 0.5$. We set the lattice constant to $a = 1$ and measure energies and temperature in the unit of t_1 . Another way to parametrize the Hamiltonian is by introducing $E_g = \varepsilon - 2(t_1 + t_2)$, which is the gap (if $E_g > 0$) or overlap (if $E_g < 0$) in the bare band structure obtained by setting the interaction strength V to zero. The chemical potential is determined by imposing

$$\frac{1}{N} \sum_i \left(\langle n_{1,i} \rangle + \langle n_{2,i} \rangle \right) = 1, \quad (2)$$

where N is the number of unit cells.

The interaction is approximated using the MF (or, equivalently, the Hartree-Fock) approximation:

$$n_{1,i} n_{2,i} \rightarrow \langle n_{1,i} \rangle n_{2,i} + \langle n_{2,i} \rangle n_{1,i} - \langle c_{2,i}^\dagger c_{1,i} \rangle c_{1,i}^\dagger c_{2,i} - \langle c_{1,i}^\dagger c_{2,i} \rangle c_{2,i}^\dagger c_{1,i}. \quad (3)$$

The first (last) two terms correspond to Hartree (Fock) contributions. Note that the Hartree terms are often omitted because they “only” shift bands and this effect is assumed

to be included in onsite energies. Moreover, it is the Fock terms which produces the excitonic insulating state. Nevertheless, we decide to keep all terms.

In this way, we obtain the MF Hamiltonian H_{MF} , which is quadratic and can be transformed to momentum space via $c_{\alpha,i} = \frac{1}{N} \sum_k e^{ikj} c_{\alpha,k}^\dagger$. In addition, we introduce the order parameter

$$\Delta = -V \langle c_{2,i}^\dagger c_{1,i} \rangle = -\frac{V}{N} \sum_k \langle c_{2,k}^\dagger c_{1,k} \rangle \quad (4)$$

and energies

$$\varepsilon_{\alpha,k} = (-1)^\alpha (-t_\alpha \cos k + \varepsilon), \quad \alpha = 1, 2. \quad (5)$$

The order parameter Δ quantifies the interband hybridization. Note that because there are no kinetic terms of the type $t_{12} c_{1i}^\dagger c_{2i} + \text{H.c.}$, a non-zero hybridization can arise only due to interaction. This is why $|\Delta| > 0$ is understood as a signature of the excitonic insulating phase. The fact that $t_{12} = 0$ also means that the model without interaction possesses the $U_1(1) \times U_2(1)$ symmetry related to the conservation of charge in each bands. As $|\Delta| > 0$, this symmetry is broken and the remaining symmetry is $U(1)$ for conservation of charge in the entire system.

The MF Hamiltonian H_{MF} can be written as

$$H_{\text{MF}}[\Delta] = \sum_k \begin{pmatrix} c_{1,k}^\dagger & c_{2,k}^\dagger \end{pmatrix} \underbrace{\begin{pmatrix} \varepsilon_{1,k} + V \langle n_2 \rangle & \Delta \\ \Delta^* & \varepsilon_{2,k} + V \langle n_1 \rangle \end{pmatrix}}_{\equiv \mathbf{h}_k} \begin{pmatrix} c_{1,k} \\ c_{2,k} \end{pmatrix}. \quad (6)$$

We solve Eq. 4 self-consistently to determine the equilibrium order parameter Δ . Diagonalizing matrices $\mathbf{h}_k$ we obtain the effective band structure:

$$H_{\text{MF}} = \sum_k (E_{+,k} d_{+,k}^\dagger d_{+,k} + E_{-,k} d_{-,k}^\dagger d_{-,k}). \quad (7)$$

The bands $E_{\pm,k}$ are determined by the unitary matrix U_k such that $\mathbf{h}_k = U_k \text{diag}(E_{-,k}, E_{+,k}) U_k^\dagger$. The corresponding operators are

$$\begin{pmatrix} d_{-,k} \\ d_{+,k} \end{pmatrix} = U_k \begin{pmatrix} c_{1,k} \\ c_{2,k} \end{pmatrix}. \quad (8)$$

The current operator j is derived from the continuity equation for particle number conservation, and can be calculated as a time-derivative of the polarization operator $P = \sum_{i,\alpha} (R_i + r_\alpha) n_{\alpha,i}$:

$$j = \frac{\partial P}{\partial t} = i[H, P] = \sum_k \begin{pmatrix} c_{1,k}^\dagger & c_{2,k}^\dagger \end{pmatrix} \underbrace{\begin{pmatrix} -2t_1 \sin k & 0 \\ 0 & 2t_2 \sin k \end{pmatrix}}_{\equiv \mathbf{J}_k} \begin{pmatrix} c_{1,k} \\ c_{2,k} \end{pmatrix} \quad (9)$$

Equivalently, the current operator is calculated as $\mathbf{J}_k = -\frac{\partial \mathbf{h}_{k-A}}{\partial A} \big|_{A=0}$

Because interaction is local, the current-current response function has no vertex corrections, in other words, it is described by the bubble

$$\chi_{jj}(\omega) = \frac{1}{N} \sum_{k,ij} |(\tilde{\mathbf{J}}_k)_{ij}|^2 \Pi_{ij,k}^{(0)}(\omega). \quad (10)$$

where the sum runs over $i, j = +, -$. Here, $\tilde{J}_k = U_k^\dagger J_k U_k$ and

$$\Pi_{ij,k}^{(0)}(\omega) = \int d\epsilon \left[A_{i,k}(\epsilon) G_{j,k}^R(\epsilon + \omega) f(\epsilon) + A_{i,k}(\epsilon + \omega) G_{j,k}^A(\epsilon) f(\epsilon + \omega) \right], \quad (11)$$

where f is the Fermi-Dirac distribution. We assume a finite broadening Γ of single-particle spectral functions $A_{i,k}$, such that the retarded (advanced) Green's function $G_{ik}^{R(A)}$ and $A_{i,k}$ are given by

$$G_{i,k}^{R(A)}(\epsilon) = \frac{1}{\epsilon - E_{i,k} + (-)i\Gamma}, \quad A_{i,k}(\epsilon) = \frac{1}{\pi} \frac{\Gamma}{(\epsilon - E_{i,k})^2 + \Gamma^2}. \quad (12)$$

The width Γ is related to the quasiparticle lifetime τ via $\tau = 1/2\Gamma$. I do not determine Γ self-consistently; I include it as an extra parameter.

The real part of optical conductivity is given by

$$\sigma(\omega) = \frac{2\pi}{G_0 a} \frac{1}{\omega} \text{Im} \chi_{jj}(\omega), \quad (13)$$

where $G_0 = h/e_0^2 = 1/(25.8 \text{ k}\Omega)$. Here, to get physical units, I restored the $\hbar$, which was previously set to 1. In 1d, σ has units of $1/\Omega\text{m}$.

DC conductivity is defined as the $\omega \rightarrow 0$ limit of optical conductivity:

$$\sigma_{\text{DC}} = \frac{2\pi}{G_0 a} \lim_{\omega \rightarrow 0} \frac{1}{\omega} \text{Im} \chi_{jj}(\omega). \quad (14)$$

3 Results

In Fig. 1(a), I show the phase diagram with the excitonic and the normal phase, depending on whether the order parameter at $T = 0$ has a non-vanishing value. I was instructed to obtain more detailed results for $V = 2$. For smaller values of interaction, the excitonic transition is shifted to lower temperatures. In Fig. 1(b), I show the typical temperature dependence of the order parameter for $V = 2$ and a range of E_g .

In Fig. 2, I show (a) entropy, (b) specific heat, (c) DC conductivity as a function of temperature, for $V = 2$ and a range of E_g . The insets (a0) and (b0) show the chemical potential and the order parameter, respectively. For parameters $E_g = -2.0, -1.8, -1.6$, there is the excitonic phase transition. It is accompanied by a discontinuity in specific heat.

In Fig. 3, I first illustrate the general effect of Γ . Larger value of Γ broadens the two peaks in optical conductivity spectrum.

In Fig. 4(a), I show the temperature dependence of the order parameter for $V = 2$ and $E_g = -2$. As temperature is increased, the order parameter is reduced and the gap (E'_g) starts closing in the effective band structure. Fig. 4(b) illustrates that this is reflected also in optical conductivity. In (b) and (c) I show results for $\Gamma/t = 0.005$ and $\Gamma/t = 0.01$. Apart from the peak at $\omega \approx E'_g$, which reflects interband transitions, there is also the Drude peak due to intraband transport. These are possible due to a finite Γ .

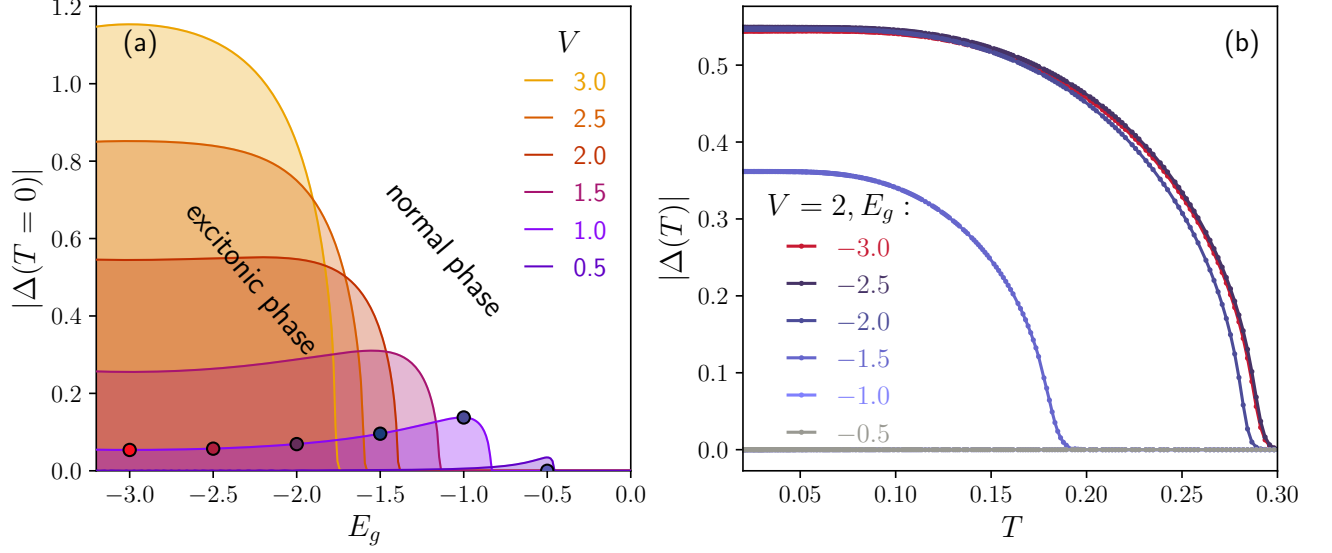


Figure 1. (a) Phase diagram determined by the value of the order parameter in the ground state. (b) Order parameter as a function of T and E_g for $V = 2$.

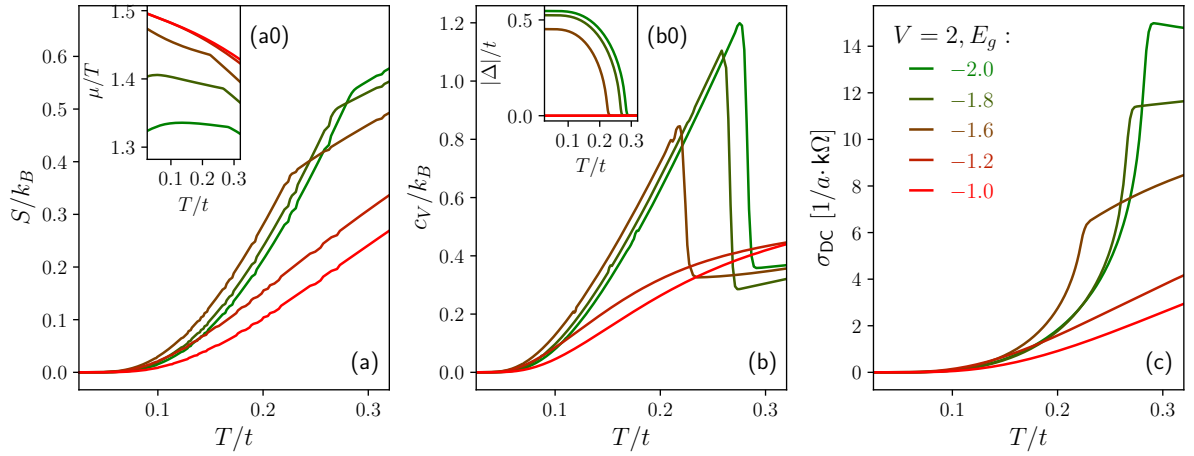


Figure 2. Entropy as a function of temperature and E_g for $V = 1$.

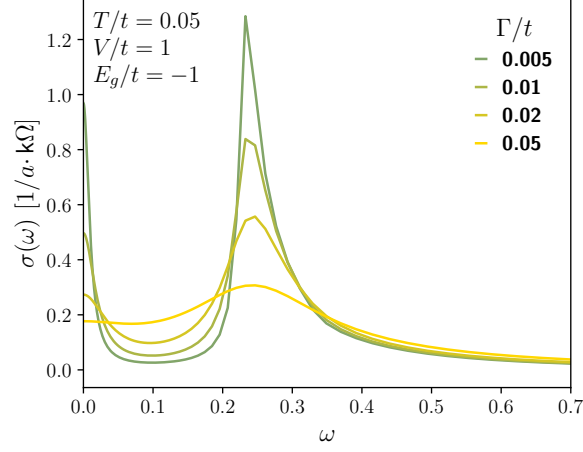


Figure 3. The effect of Γ on optical conductivity, example of $V = 1, E_g = -1, T = 0.05$ in units of t .

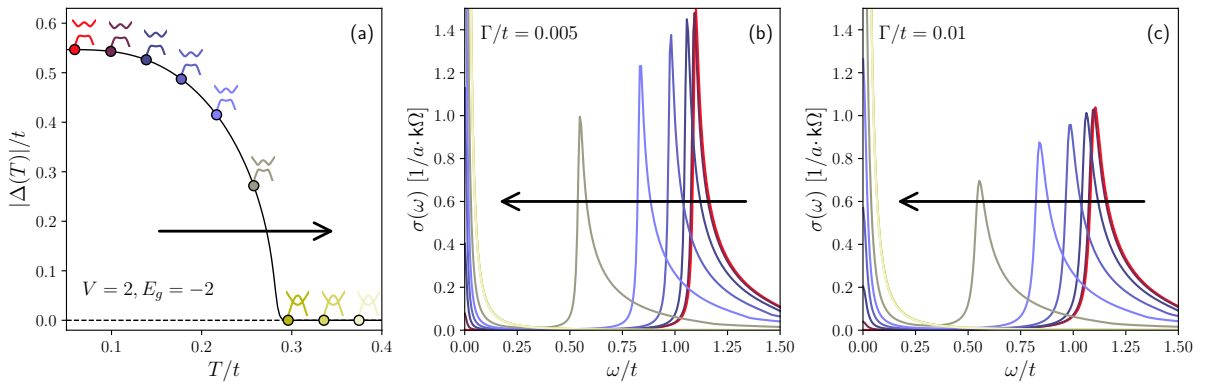


Figure 4. Parameters $V = 2$ and $E_g = -2$. (a) Temperature dependence of the order parameter and the band structure. (b), (c) Optical conductivity for $\Gamma/t = 0.005$ and $\Gamma/t = 0.01$, respectively. Arrows go from low to high temperature.